JAVA & DBMS PROJECT

Rajalakshmi Engineering College

ImpactHub .

An integrated Java OOP & Database Management System (DBMS) for NGO volunteer coordination and activity tracking.

TEAM MEMBERS & REGISTER NUMBERS

Nidharsha R (2116251001163) • Nithya Sree S K (2116251001174) • Pavithra R R (2116251001179)

Project Background & Motivation

Problem Statement & Solution Approach

MANUAL DRAWBACKS

The Manual Record Crisis

- ✕ Lost & misplaced physical registers
- ✕ Difficulty tracking volunteer activity history

High paper dependency & administrative overhead

DBMS + JAVA SOLUTION

ImpactHub Centralization

- ✓ Persistent MySQL relational data tables
- ✓ Fast Java JDBC queries with foreign key integrity
- ✓ Instant search, listing, and cancellation features

Automated record lookup & data persistence

Architecture & Features

Detailed Module Breakdown

1. Volunteer Module

Manages personal and contact details for all participating volunteers.

Attributes: Name, Phone, Email
Operations: Register, View Roster

Auto-Increment Volunteer ID

2. Activity Module

Tracks social programs such as food drives and environmental campaigns.

Attributes: Name, Location, Status
Operations: Add Event, View List

Location & Open Status Tracking

3. Registration Module

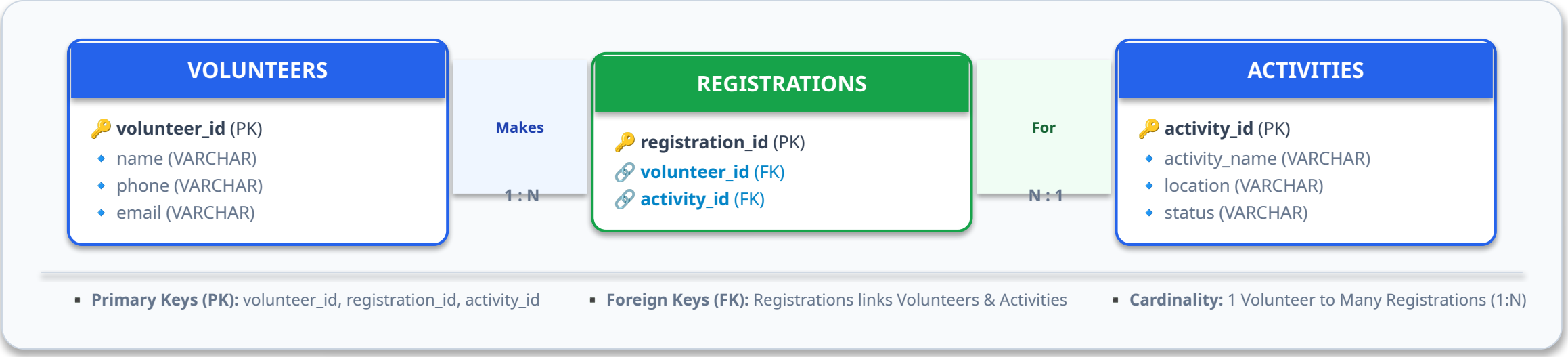
Maps volunteer participation to specific NGO events dynamically.

Foreign Keys: volunteer_id, activity_id
Operations: Link Event, Cancel

Relational Foreign Key Integrity

Database Architecture

Entity-Relationship (ER) Diagram



- **Primary Keys (PK):** volunteer_id, registration_id, activity_id
- **Foreign Keys (FK):** Registrations links Volunteers & Activities
- **Cardinality:** 1 Volunteer to Many Registrations (1:N)

Database Implementation

MySQL Table Schemas

01 DATABASE & VOLUNTEERS

```
CREATE DATABASE ngo_management;
USE ngo_management;

CREATE TABLE Volunteers (
    volunteer_id INT PRIMARY KEY
        AUTO_INCREMENT,
    name  VARCHAR(50),
    phone VARCHAR(15),
    email VARCHAR(100)
);
```

02 ACTIVITIES & REGISTRATIONS

```
CREATE TABLE Activities (
    activity_id INT PRIMARY KEY
        AUTO_INCREMENT,
    activity_name VARCHAR(100),
    location VARCHAR(100),
    status VARCHAR(20)
);

CREATE TABLE Registrations (
    registration_id INT PRIMARY KEY
        AUTO_INCREMENT,
    volunteer_id INT,
    activity_id INT,
    FOREIGN KEY (volunteer_id)
        REFERENCES Volunteers(volunteer_id),
    FOREIGN KEY (activity_id)
        REFERENCES Activities(activity_id)
);
```

Backend Connectivity

JDBC Connection & Query Flow

DBConnection.java | JDBC CONNECTION

```
public class DBConnection {

    public static Connection getConnection() {
        Connection con = null;

        try {
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/ngo_management",
                "root",
                "YOUR_PASSWORD"
            );
        } catch (Exception e) {
            System.out.println(e);
        }

        return con;
    }
}
```

1. Driver Loading

MySQL Connector/J connects the Java application to the local database.

2. Statement Execution

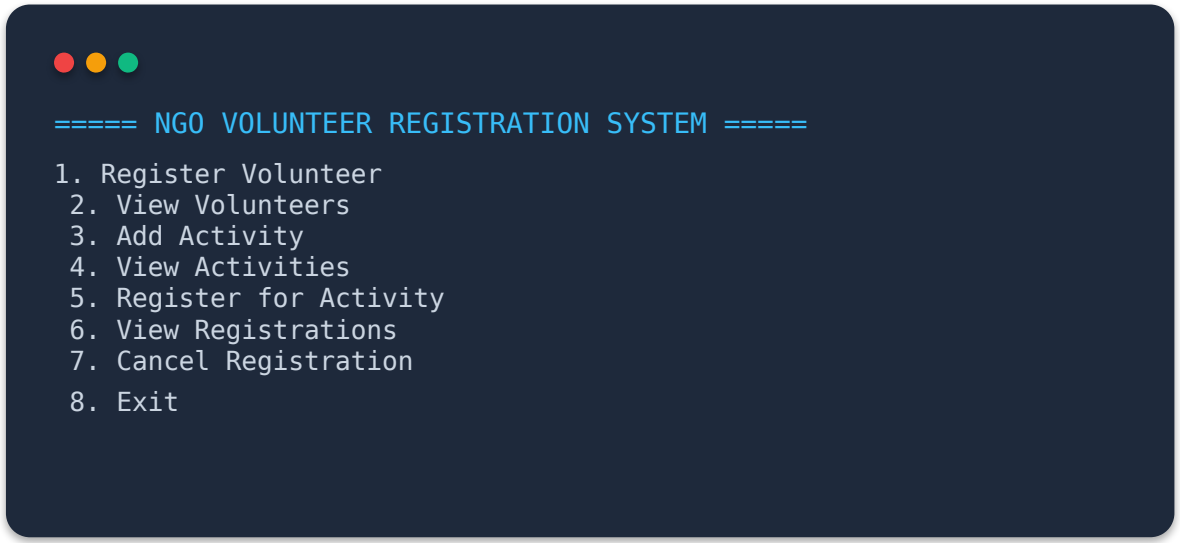
SQL statements perform database insert, update and retrieval operations.

3. ResultSet Extraction

ResultSet reads returned rows for display in the application.

System Execution

Menu Navigation & Operations



Continuous Loop: Scanner with while-switch

Keeps menu operational until option 8 is chosen.

Data Integrity: Exception Handling

Traps SQL errors if invalid foreign key IDs are entered.

Resource Safety: Connection Management

Explicitly closes Statements and ResultSets after execution.



Thank You!

ImpactHub Project • Java OOP & DBMS Presentation

Rajalakshmi Engineering College, Thandalam